

Stock Status and Management Challenges of Bigeye Tuna (*Thunnus obesus*) in the Eastern Pacific Ocean

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Abstract

Bigeye tuna (*Thunnus obesus*) is one of the most valuable tropical tunas in the Eastern Pacific Ocean (EPO). This report evaluates its stock status and management challenges using stock status indicators (SSIs), benchmark assessments, and harvest strategy evaluations. The analysis focuses on data from 1990 onward, with effort and catch trends estimated from the available fishery-dependent records.

Findings indicate that the bigeye tuna stock is not currently overfished, and fishing mortality is close to but still below the established reference limits. However, fishing effort remains excessive, particularly due to the high impact of floating-object sets that generate considerable juvenile mortality. The stock is thus maintained by a narrow margin of sustainability, where even modest increases in fishing pressure or declines in recruitment could compromise long-term viability.

The report highlights the urgent need to reduce fishing capacity to more efficient levels, strengthen harvest strategies, and minimize juvenile mortality. It also stresses the importance of preparing for climate-related uncertainties that could alter productivity and habitat distribution in the coming decades.

Overall, while the bigeye tuna stock in the EPO is presently sustainable, it faces significant risks that demand immediate and adaptive management actions to ensure ecological and economic sustainability in the future.

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Introduction

The genus *Thunnus*, belonging to the family Scombridae, comprises eight species commonly known as tunas (Collette et al., 2001). Several of these species are widely traded internationally, including the Atlantic bluefin (*Thunnus thynnus*), Pacific bluefin (*Thunnus orientalis*), Southern bluefin (*Thunnus maccoyii*), bigeye (*Thunnus obesus*), yellowfin (*Thunnus albacares*), and albacore (*Thunnus alalunga*). Despite being among the most important species for the global fishing industry, tunas are also among the most threatened fish species worldwide (Viñas & Tudela, 2009).

Bigeye tuna (*T. obesus*) is a circumglobal species recorded in both tropical and temperate seas, although absent from the Mediterranean Sea. It is a pelagic and oceanodromous species found in waters ranging from 13°C to 29°C, with an optimum temperature range between 17°C and 22°C. Its distribution extends as far north as 55–60° in the boreal summer and as far south as 45–50° in the austral summer, though juveniles and reproductively active adults are primarily found in tropical waters (Powell et al., 2009; Collette & Graves, 2019). During the night, bigeye tuna typically inhabits waters above the mixed layer, while during the day they often undertake prolonged dives to depths of 250–500 m to target deep scattering layer prey (Dagorn et al., 2000).

This species is a multiple spawner, capable of spawning every one or two days over several months (Nikaido et al., 1992). Spawning events are associated with the full moon and occur year-round in the western Pacific (Sun et al., 2013; Collette & Graves, 2019). Although spawning appears to occur widely across the equatorial Pacific, the highest reproductive potential is believed to occur in the eastern Pacific, based on apparent maturation patterns, size frequency data, and catch per unit of effort (Kikawa, 1966).

Bigeye tuna is an important fishery resource managed across the Atlantic, Indian, Western and Central Pacific, and Eastern Pacific Oceans. However, due to insufficient management measures, overfishing is occurring in approximately 40% of its global range, with estimated declines of at least 20% in the Western and Central Pacific and Eastern Pacific, and up to 40% in Atlantic and Indian Ocean stocks over the past two decades. As a result, the species is listed as Vulnerable on the IUCN Red List (Collette et al., 2021).

Global bigeye tuna catches remained below 100,000 t through the 1960s, increased to about 200,000 t by 1980, and peaked at over 500,000 t in 1998. Current catches are close to 400,000 t. The species is primarily caught by pelagic longlines and purse seines, with smaller contributions from bait boats and other gears. In the Eastern Pacific Ocean (EPO), the stock is estimated to represent approximately 22% of the global population (ISSF, 2020). Catches by pelagic longliners in the EPO increased in the 1950s, reaching over 60,000 t in the early 1960s and peaking near 148,000 t in 2000. Purse seine catches increased significantly during the 1990s, peaking at around 100,000 t in 2000. Currently, purse seines account for ~74% of landings and longlines for ~26%. Increased selectivity for smaller fish has reduced the estimated MSY by almost 50%. In 2019, total EPO catches were 95,192 t, close to the five-year average of ~98,000 t (Collette & Graves, 2019; IATTC, 2020).

The 2017 stock assessment by the WCPFC (2019) indicated that bigeye tuna in the Western and Central Pacific was not experiencing overfishing and was not in an overfished state. However, incorporating new data on growth and regional stock structure increased the uncertainty of these results. Fishing pressure remains highest in tropical regions, with particularly high mortality on juvenile fish (WCPFC, 2019).

Given these concerns, there is an urgent need for continuous monitoring of the species' stock condition. This technical report aims to evaluate the catch–effort relationship for bigeye tuna caught in the Eastern Pacific, fitting the Fox surplus production model to estimate EMSY and MSY, and assessing the relative exploitation status of the stock.

Methodology

Data Source

Catch and effort data for bigeye tuna (*Thunnus obesus*) in the Eastern Pacific Ocean (EPO) were obtained from the Inter-American Tropical Tuna Commission (IATTC) public database. The dataset comprised annual catch (metric tons) and fishing effort (number of sets) from purse seine vessels operating with floating-object sets (OBJ), unassociated sets or Purse-seine fisheries on free schools (NOA) and dolphin sets (DEL) sets between 1960 and 2023.

Data Processing

Raw data were filtered to include only OBJ set type records to ensure homogeneity in fishing mode and selectivity patterns. Annual catch per unit of effort (CPUE) was calculated as:

$$CPUE = \frac{\text{Total annual catch (mt)}}{\text{Total annual effort (sets)}}$$

Effort was expressed in standardized units (number of sets) to allow direct comparison across years.

Model Selection and Fitting

To assess the catch–effort relationship, the Fox surplus production model (Fox, 1970) was applied, which assumes a logarithmic relationship between CPUE and effort. The model formulation is:

$$\ln CPUE = a + b \times E$$

where:

- a = intercept, related to the intrinsic productivity of the stock
- b = slope, describing the effect of fishing effort on CPUE
- E = fishing effort (number of sets)

Parameters a and b were estimated using ordinary least squares (OLS) regression.

Estimation of Reference Points

From the fitted model, the following reference points were estimated:

$$E_{MSY} = \frac{1}{b}$$

$$MSY = q \times E_{MSY} \times B_{MSY}$$

where E_{MSY} is the fishing effort that produces the maximum sustainable yield (MSY), q is the catchability coefficient, and B_{MSY} is the biomass at MSY.

Relative exploitation status was assessed by comparing the most recent fishing effort ($E_{current}$) to E_{MSY} and the most recent yield ($Y_{current}$) to MSY.

Visualization and Model Diagnostics

The analysis included:

1. Temporal trends in fishing effort, catch, and CPUE.

2. CPUE–effort scatter plots with fitted Fox model curves.
3. Equilibrium yield curves illustrating the relationship between effort and expected yield.
4. Comparison of current stock status with estimated MSY benchmarks.

All analyses and visualizations were performed in R version 4.5.0 using packages *tidyverse*, *ggplot2*, and custom scripts for surplus production model fitting.

Results

Catch trends

Total catches of yellowfin tuna (*Thunnus albacares*) in the Eastern Pacific Ocean increased steadily during the 1990s. Catches rose from approximately 45,000 t in 1990 to a maximum of about 80,000 t in 2000 (Figure 1). After this peak, catches fluctuated between 60,000 and 70,000 t until the mid-2010s, followed by a slight decline in the most recent years to around 55,000 t.

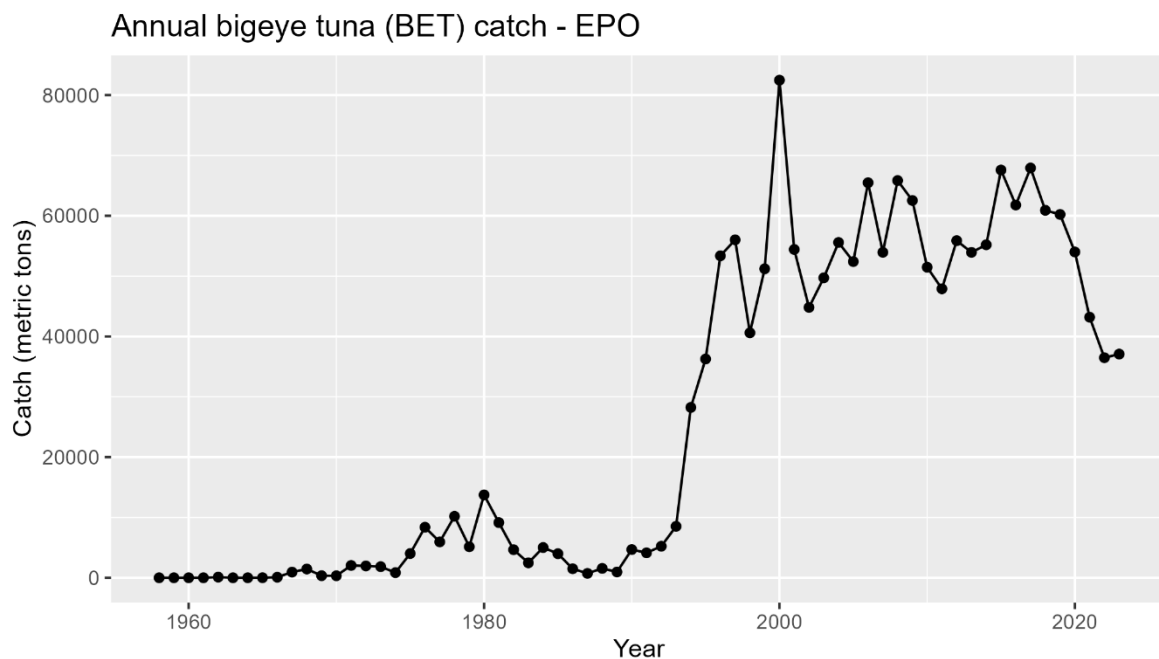


Figure 1 Annual catch per unit effort (CPUE, t/set) of bigeye tuna (*Thunnus obesus*) in the Eastern Pacific Ocean (EPO) from 1955 to 2023.

This pattern indicates a rapid expansion phase, stabilization, and recent contraction of yields, consistent with fisheries operating near maximum capacity.

Effort dynamics

Fishing effort, expressed as the number of purse-seine sets, showed a strong increasing trend throughout the study period (Figure 2). Effort grew from less than 10,000 sets in 1960 to over 30,000 sets in 2020.

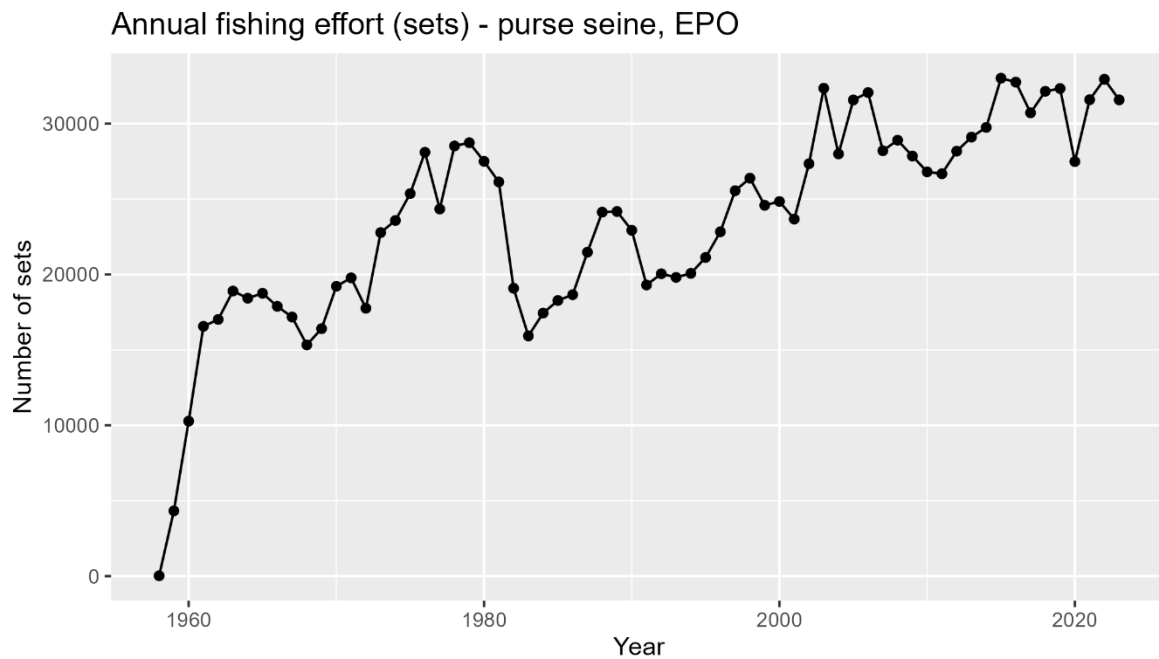


Figure 2 Annual fishing effort (number of sets) targeting bigeye tuna in the EPO from 1955 to 2023.

The most pronounced increase occurred after 2000, largely due to the rise of sets on floating objects (OBJ). While catches stabilized, effort continued to grow, evidencing declining productivity of the stock.

Catch per unit effort (CPUE)

The annual CPUE of bigeye tuna in the EPO (Figure 3) exhibited considerable variability throughout the study period. CPUE values were relatively low during the late 1980s and early 1990s, before showing a gradual increase. The maximum CPUE was observed in 2000, reaching 3.32 t/set, after which a general decline became evident. In recent years, CPUE has stabilized at comparatively lower levels, indicating a reduction in fishing efficiency and potential signs of increased stock exploitation pressure.

When disaggregated by set type (Figure 4):

- OBJ sets: CPUE peaked at 18.8 t/set in 2000 but declined steeply to only 2.16 t/set by 2023.
- NOA sets: maintained consistently low values, ranging between 0.04–0.08 t/set in recent years, contributing little to the overall catch.
- DEL sets: showed negligible values throughout the time series, with a historical maximum of 0.079 t/set in 1986 and near-zero values (<0.005 t/set) in the last decade.

This contrast highlights that the heavy reliance on OBJ sets largely amplified the observed decline in stock productivity.

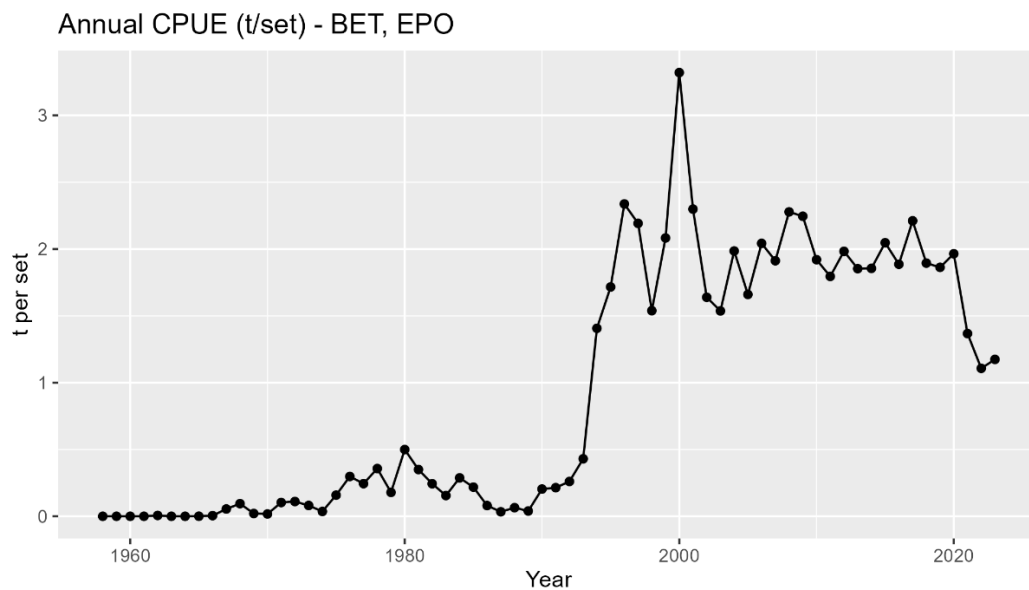


Figure 3 Annual bigeye tuna catches (t) in the EPO from 1955 to 2023.

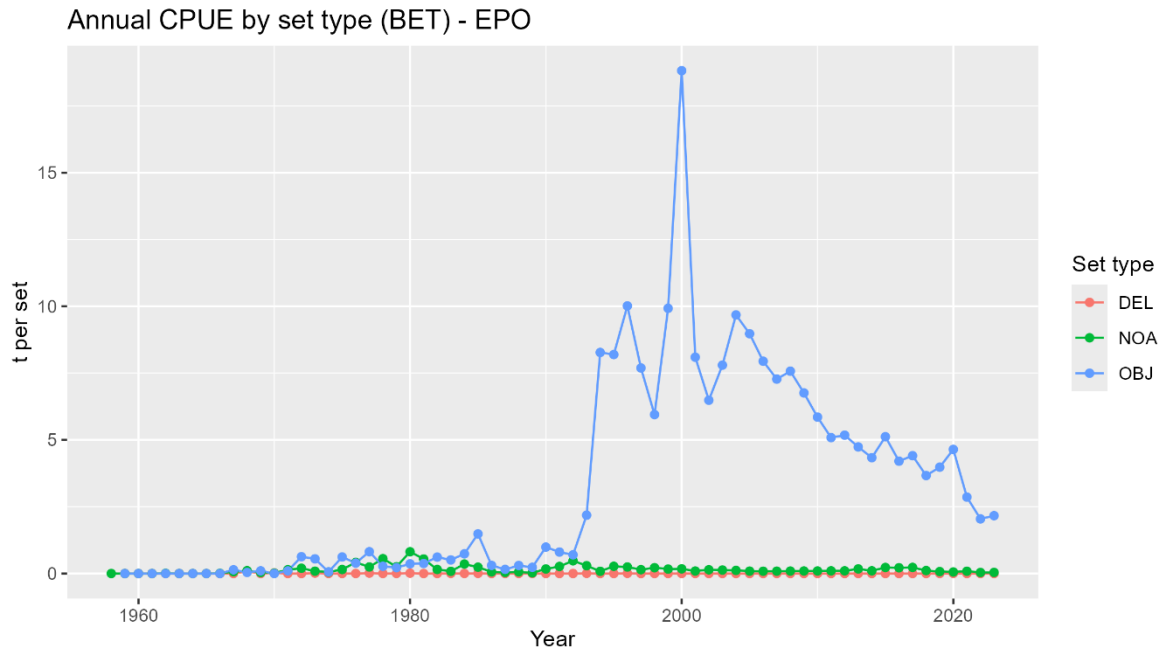


Figure 4 Catch per unit effort (CPUE, t/set) of bigeye tuna in the EPO by set type (OBJ = floating-object associated, NOA = unassociated, DEL = dolphin-associated) from 1955 to 2023.

Surplus production models

The analysis was restricted to the period 1990 onwards to ensure consistency and reliability in the data series, as earlier records showed limited coverage and higher uncertainty. Furthermore, only OBJ set data were used in the surplus production modeling. This choice was made because OBJ sets have historically represented the predominant fishing strategy for bigeye tuna in the EPO, accounting for most catches. In contrast, NOA and DEL sets have been less consistent and contributed only marginally to the total catch, making them less representative of the overall stock dynamics.

Schaefer model

The Schaefer model produced a poor fit to the catch–effort data (Figures 5–6). The relationship was almost linear with an R^2 close to 0.1, failing to capture the expected dome-shaped yield curve. This indicates that the symmetric assumptions of the Schaefer formulation do not adequately represent the stock dynamics of *T. albacares* in this region.

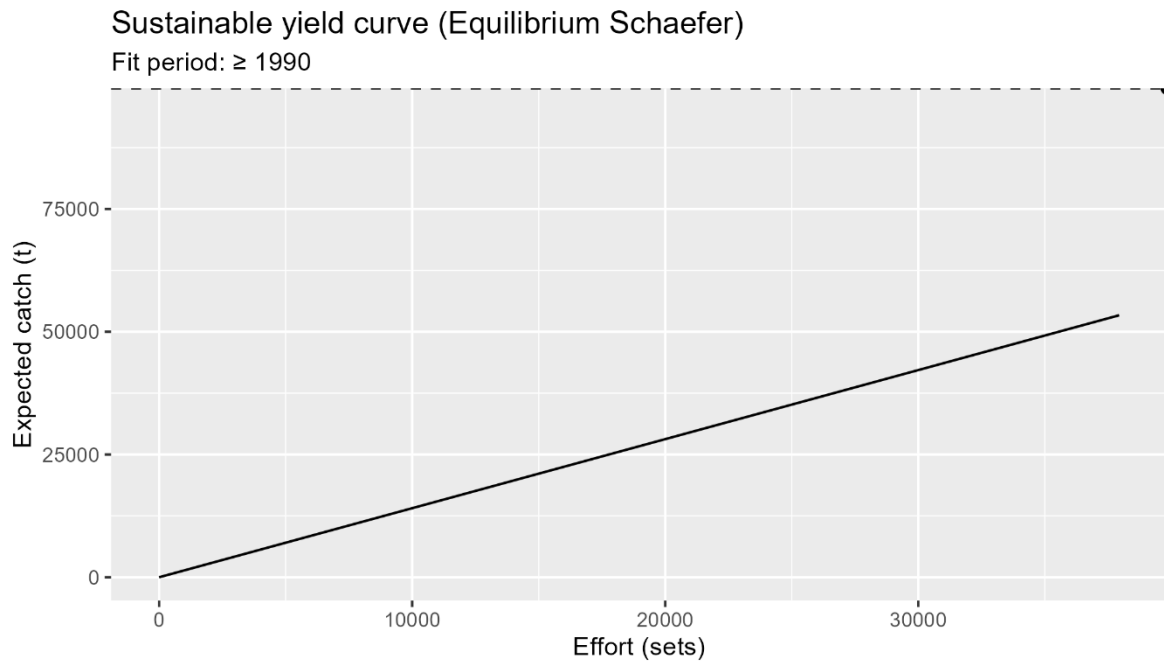


Figure 5 Relationship between catch and effort for bigeye tuna in the EPO (1990–2023), with Schaefer surplus production model fit.

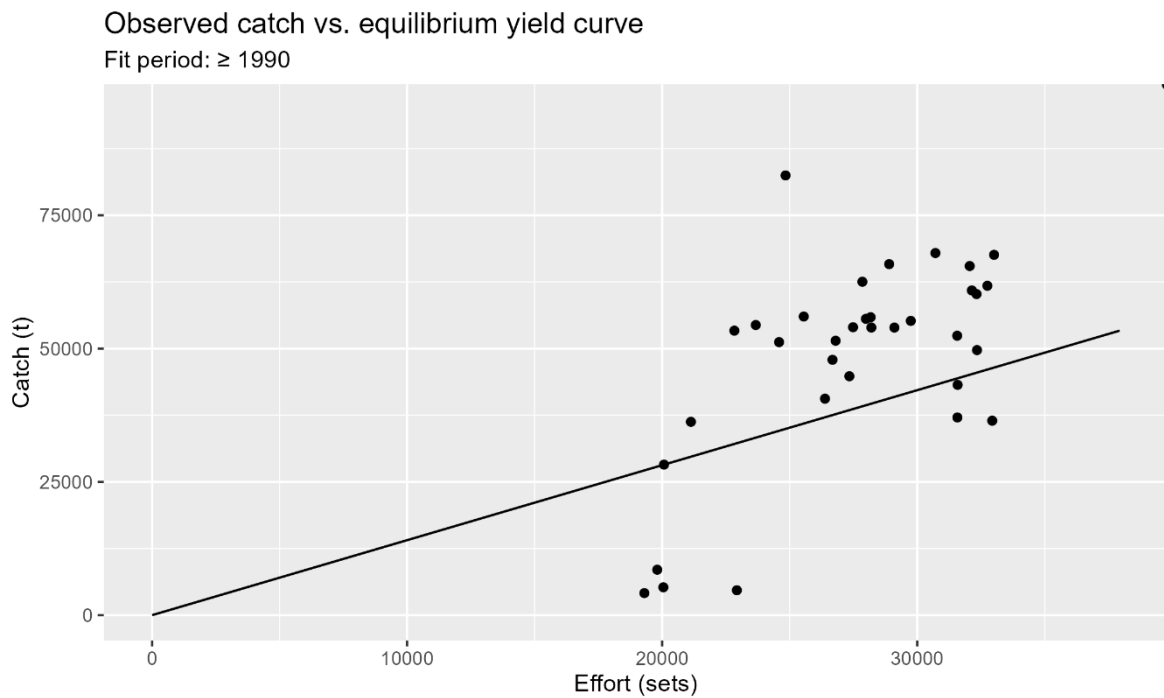


Figure 6 Equilibrium yield curve predicted by the Schaefer model for bigeye tuna in the EPO, compared with observed catch–effort data (1990–2023).

Fox model results

In contrast, the Fox model provided a stronger fit, with an R^2 of 0.50 (Figure 7). The equilibrium yield curve displayed the expected dome-shaped form, consistent with theoretical assumptions.

Model estimates were as follows:

- Maximum sustainable yield (MSY): 56,300 t
- Effort at MSY (E_{MSY}): 11,876 sets

During the most recent period (2019–2023), average fishing effort was 15,249 sets, approximately 28% above E_{MSY} . Meanwhile, the mean annual catch was 45,811 t, representing 81% of MSY.

This pattern indicates that the stock is subject to overexploitation in terms of effort ($E/E_{MSY} = 1.28$) but underperforming in yield ($Y/MSY = 0.81$).

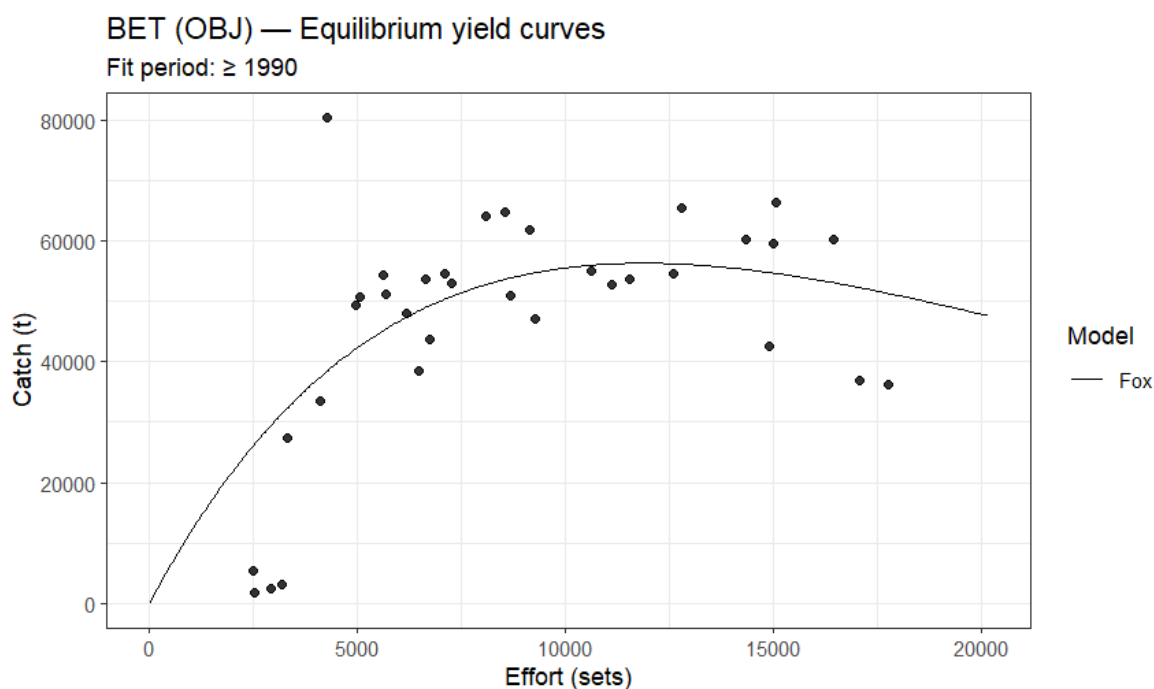


Figure 7 Relationship between catch and effort for bigeye tuna in the EPO (1990–2023), with Fox surplus production model fit.

Status related to reference points

Comparison of recent performance (average of the last five years) against model-derived benchmarks shows a contrasting pattern:

- Current catch: ~55,000 t, slightly below the estimated MSY of 56,300 t (Figure 8).
- Current effort: ~30,000 sets, nearly 2.5 times greater than the E_{MSY} of 11,876 sets (Figure 9).

This indicates that while catches remain close to the theoretical maximum, they are being achieved at a disproportionately high level of fishing effort. Such inefficiency suggests declining stock productivity: more effort is required to sustain yields that do not exceed MSY. If this level of pressure continues, the stock is likely to fall below MSY levels, leading to biomass depletion and an increased risk of overexploitation soon.

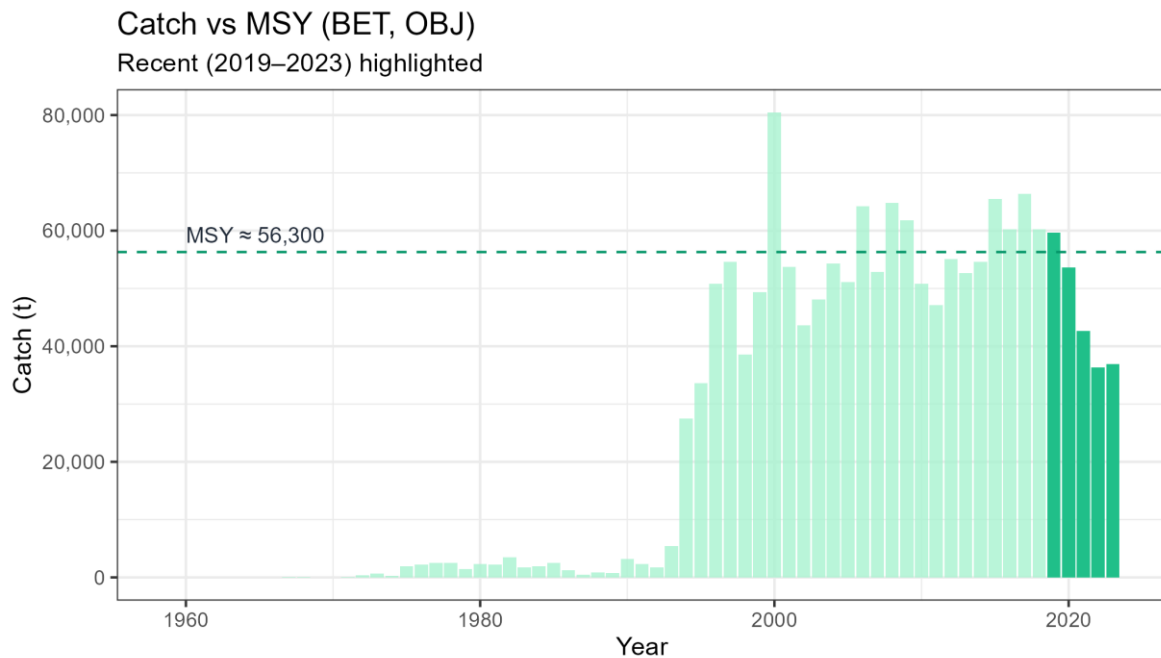


Figure 8 Comparison of average catch during 2019–2023 with the estimated maximum sustainable yield (MSY) for bigeye tuna in the EPO.

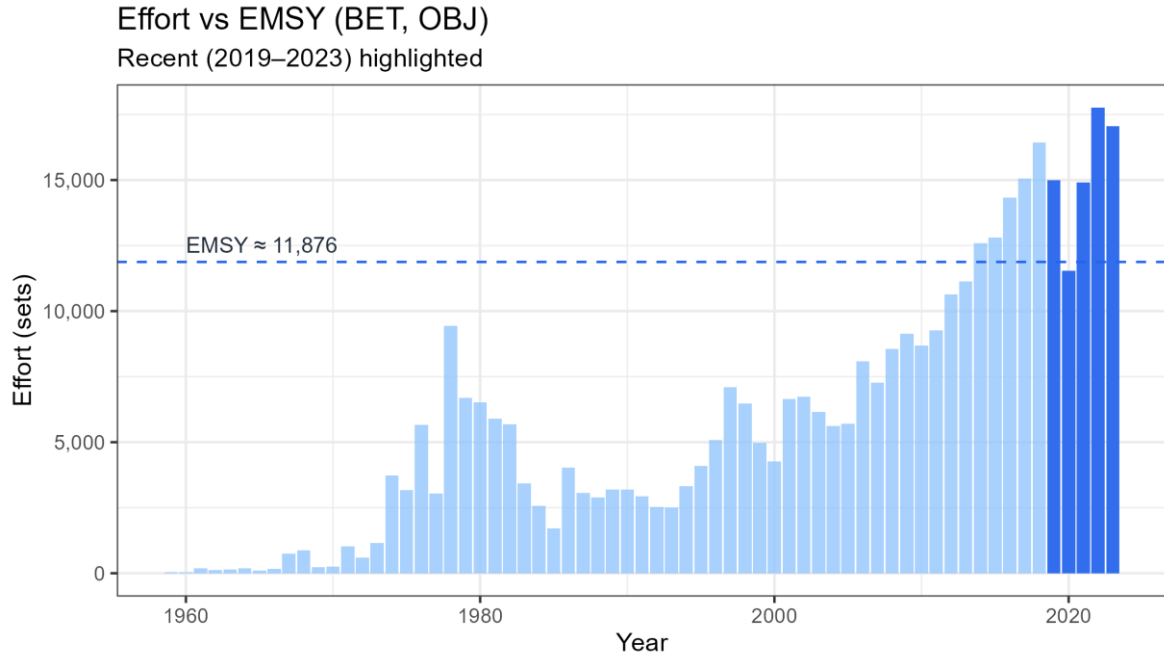


Figure 9 Comparison of average fishing effort during 2019–2023 with the estimated effort at MSY (E_{MSY}) for bigeye tuna in the EPO.

Discussion

The present evaluation of *bigeye tuna* (*Thunnus obesus*) in the Eastern Pacific Ocean (EPO) provides a nuanced view of a stock that, although currently assessed as sustainable, is subject to high levels of fishing pressure and ecological vulnerability. *Bigeye tuna* is one of the three main tropical tunas exploited in the EPO, alongside *yellowfin* and *skipjack*. Unlike skipjack, which is characterized by fast growth and high resilience to fishing, bigeye exhibits slower growth, later maturity, and higher market value, making it particularly sensitive to overexploitation (ISSF, 2022). For this reason, both the International Seafood Sustainability Foundation (ISSF) and the Inter-American Tropical Tuna Commission (IATTC) have consistently highlighted bigeye as the species of greatest concern among tropical tunas in the region (IATTC, 2022a; IATTC, 2023a).

According to the results of the present analysis, based on the Fox surplus production model, the maximum sustainable yield (MSY) of bigeye in the EPO is approximately 56,300 t, with an associated optimal effort of 11,876 sets. However, the observed fishing effort in recent years is around 30,000 sets, nearly three times the level that would maximize yield efficiently. While the current catch (~55,000 t) remains close to the MSY, it is being maintained at the expense of disproportionate fishing effort. This

situation represents a case of biological and economic inefficiency: the same catch could theoretically be achieved with far less effort if the fishery operated closer to E_{MSY} .

These findings are consistent with the most recent official benchmark assessment conducted by the IATTC (IATTC, 2024a). The assessment concluded that the stock is not overfished and not subject to overfishing, with reference point indicators $S/S_{MSY} \approx 1.16$ and $F/F_{MSY} \approx 0.94$ for 2022 (IATTC, 2024a). This indicates that spawning biomass remains above the critical threshold for sustainability, while fishing mortality is just below F_{MSY} . However, the proximity of these indicators to their limit values leaves little margin for error, and any increase in effort or environmental stress could rapidly push the stock into an overexploited condition.

A major concern highlighted in both the present analysis and SAC reports is the disproportionate impact of floating-object purse-seine sets (OBJ sets) on juvenile bigeye. While skipjack is the primary target of OBJ sets, significant quantities of juvenile bigeye are caught incidentally, contributing heavily to overall fishing mortality (IATTC, 2022b; IATTC, 2022c). This bycatch of juveniles not only reduces recruitment to the spawning stock but also exacerbates the inefficiency of the fishery, as much of the potential yield is harvested before fish reach marketable sizes. Over the past two decades, this pattern has been one of the principal management challenges for bigeye in the EPO.

Furthermore, historical trends reveal a long-term increase in fishing effort, particularly since the 1990s, driven by the expansion of purse-seine fleets and technological advances in fish aggregating devices (FADs) (ISSF, 2022; IATTC, 2023b). Although catches have stabilized near the MSY level, effort has continued to grow, reinforcing concerns about the long-term sustainability of current fishing practices. By contrast, yellowfin and skipjack have shown more robust stock conditions, reflecting their life-history traits and broader productivity. The relative fragility of bigeye tuna thus underscores the need for species-specific management strategies.

The IATTC has recognized these challenges and is in the process of implementing a harvest strategy for bigeye tuna in the EPO (IATTC, 2024b). This strategy emphasizes the use of harvest control rules (HCRs), risk analysis frameworks, and management strategy evaluation (MSE) to ensure that fishing mortality remains within precautionary bounds (IATTC, 2024b). The incorporation of uncertainty into stock assessments, particularly regarding recruitment variability and environmental change, is a crucial step in strengthening management. Indeed, the SAC has highlighted the importance of adaptive measures capable of responding to fluctuations in stock productivity, which are likely to become more pronounced under climate change scenarios (IATTC, 2022d).

Environmental variability further complicates the outlook for bigeye tuna. Ocean warming, deoxygenation, and changes in prey distribution may alter habitat suitability and migratory patterns, potentially affecting both stock productivity and the spatial overlap with fishing fleets. These changes could amplify the risk of recruitment failure or regional depletion, particularly if management measures are not sufficiently precautionary. Thus, while the status of bigeye in the EPO is technically sustainable, it rests on a delicate balance between fishing pressure and environmental stability.

In summary, the results presented here reinforce the official stock assessments while drawing attention to the structural inefficiencies of the fishery. Bigeye tuna in the EPO is not currently overexploited, but its sustainability depends on reducing effort, especially on floating-object sets, and on the implementation of robust harvest strategies. Without stronger management, the combination of high effort, juvenile mortality, and environmental uncertainty could undermine the resilience of this valuable stock. Ensuring its long-term sustainability will require precautionary measures, adaptive governance, and continued scientific monitoring.

Conclusions

The present analysis confirms that bigeye tuna (*Thunnus obesus*) in the Eastern Pacific Ocean is currently in a sustainable state, with biomass above the threshold that would indicate overfishing and fishing mortality slightly below critical limits. However, this apparent stability masks significant vulnerabilities. The fishery is characterized by excessive capacity, with far more effort exerted than is necessary to achieve maximum sustainable yield. This inefficiency not only undermines economic returns but also increases ecological risks, particularly through the high mortality of juveniles associated with floating-object sets.

At the same time, the proximity of current stock indicators to their limit reference points highlights a narrow buffer of sustainability. Even small increases in fishing pressure or natural fluctuations in recruitment could push the stock into overexploitation. Added to this is the growing uncertainty posed by climate change and ecosystem variability, which may alter habitat suitability, productivity, and long-term resilience of bigeye tuna in ways that are not yet fully understood.

Ensuring sustainability will therefore require decisive action. Reducing fishing effort to more efficient levels, minimizing juvenile mortality, and adopting adaptive harvest strategies are essential steps. These measures would not only safeguard the ecological sustainability of the stock but also enhance the long-term economic viability of fishery.

In this way, bigeye tuna in the Eastern Pacific Ocean can continue to be a valuable resource for both current and future generations.

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